Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Edwin H. Smith and Sons Dairy

Physical address of dairy:

4836 N Quincy RD	Denair	Stanislaus	95316
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X057-X009-XXXX X045-X057-X019-XXXX

B. OPERATORS

Edwin H. Smith and Sons

Operator name: Edwin H. Smith and Sons

Telephone no.: (123) 000-0000 (Landline) (Cellular)

4905 N Quincy RD	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Edwin H. Smith and Sons

Legal owner name: Edwin H. Smith and Sons

Telephone no.: (123) 000-0000 (Landline) (Cellular)

4905 N Quincy RD	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	22	69	36	18	21
Number under roof	179	0	0	0	0	0
Maximum number	179	22	69	36	18	21
Average number	179	22	69	36	18	21
Avg live weight (lbs)	1,600	1,550	975	340		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 6,028.75 *tons per reporting period*

Total nitrogen from manure: 75,456.10 *lbs per reporting period*

After ammonia losses (30% loss applied): 52,819.27 *lbs per reporting period*

Total phosphorus from manure: 12,419.43 *lbs per reporting period*

Total potassium from manure: 34,044.95 *lbs per reporting period*

Total salt from manure: 89,341.05 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 15,265,468 *gallons*

Total nitrogen generated: 5,070.13 *lbs*

Total phosphorus generated: 1,230.59 *lbs*

Total potassium generated: 5,235.74 *lbs*

Total salt generated: 53,503.94 *lbs*

	15,265,468	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	15,265,468	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal Water	Surface water
Domestic Well/ #1 5106 N Quincy	Ground water
Domestic Well/ #11 Sperry's	Ground water
Domestic Well/ #2 Paul 5200 N Quincy	Ground water
Domestic Well/ #27 Hyline	Ground water

Source Description	Type
Domestic Well/ #5 Nutchers Dairy	Ground water
Domestic Well/ 16 Home Dairy	Ground water
Domestic Well/ 2519 Barn Hart Rd	Ground water
Domestic Well/ 5818 S Gratton Rd	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1-1	14	14	2	both	X045-X062-X010-XXXX
1-10	18	18	0	none	X045-X061-X010-XXXX
1-11	17	17	0	none	X045-X039-X014-XXXX
1-2	58	58	0	none	X045-X052-X013-XXXX
1-5	78	78	0	none	X045-X059-X005-XXXX
1-7	24	24	0	none	X045-X010-X009-XXXX
1-9	9	9	0	none	X045-X061-X022-XXXX
2-1	12	12	1	manure	X019-X036-X032-XXXX
2-10	52	52	0	none	X045-X057-X019-XXXX
2-11	60	60	0	none	X045-X057-X009-XXXX
2-12A	40	40	0	none	X045-X056-X004-XXXX
2-12B	37	37	0	none	X045-X056-X004-XXXX
2-13	80	80	1	manure	X045-X057-X017-XXXX
2-14	37	37	0	none	X019-X032-X010-XXXX
2-15	18	18	0	none	X019-X034-X020-XXXX
2-16	38	38	0	none	X019-X034-X003-XXXX
2-17	38	38	0	none	X019-X033-X024-XXXX
2-18	65	65	0	none	X019-X025-X033-XXXX X019-X025-X041-XXXX
2-2	9	9	0	none	X019-X036-X029-XXXX X019-X036-X031-XXXX X019-X036-X050-XXXX
2-3	20	20	0	none	X019-X035-X011-XXXX
2-30	9	9	0	none	X019-X035-X004-XXXX
2-31	15	15	1	manure	X019-X032-X009-XXXX
2-4	19	19	0	none	X019-X035-X010-XXXX
2-5	8	8	0	none	X019-X036-X020-XXXX
2-6	20	20	0	none	X019-X036-X052-XXXX X045-X057-X004-XXXX
2-7	30	30	0	none	X045-X057-X017-XXXX
2-8	10	10	0	none	X045-X057-X015-XXXX
2-9	18	18	2	both	X045-X057-X008-XXXX
3-1	40	40	1	manure	X073-X001-X003-XXXX
3-11	20	20	0	none	X073-X011-X002-XXXX

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
3-13	20	20	0	none	X073-X010-X005-XXXX
3-15	15	15	0	none	X024-X012-X009-XXXX
3-16	6	6	2	both	X024-X012-X005-XXXX
3-17	18	18	2	both	X024-X036-X003-XXXX
3-18	9	9	0	none	X073-X016-X007-XXXX
3-19	8	8	0	none	X024-X037-X023-XXXX
3-2	47	47	1	manure	X073-X002-X001-XXXX
3-20	28	28	3	both	X051-X003-X011-XXXX
3-21	61	61	1	manure	X024-X012-X001-XXXX
3-33	17	17	2	both	X073-X007-X014-XXXX X073-X007-X015-XXXX
3-34	40	40	0	none	X073-X006-X001-XXXX
3-35	39	39	0	none	X073-X002-X002-XXXX
3-36	42	42	0	none	X024-X012-X015-XXXX
3-38	42	42	0	none	X240-X012-X015-XXXX
3-7	20	20	0	none	X024-X012-X014-XXXX
3-8	8	8	1	both	X073-X005-X011-XXXX
3-9	13	13	1	both	X073-X005-X009-XXXX
Nutcher's	80	80	0	none	X045-X057-X014-XXXX
Totals for areas that were used for application	377	377	21		
Totals for areas that were not used for application	1,049	1,049	0		
Land application area totals	1,426	1,426	21		

B. CROPS AND HARVESTS

1-1

Field name: 1-1

1-1

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	231.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.50	0.00	0.00	0.00	0.00

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	380.31 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.17	0.00	0.00	0.00	0.00

2-1

Field name: 2-1

09/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 09/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	105.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.75	0.00	0.00	0.00	0.00

2-13

Field name: 2-13

09/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 09/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	766.50 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.58	0.00	0.00	0.00	0.00

2-31

Field name: 2-31

09/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 09/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	144.38 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.63	0.00	0.00	0.00	0.00

2-9

Field name: 2-9

2-9

09/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 09/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	267.75 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.88	0.00	0.00	0.00	0.00

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	823.04 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	45.72	0.00	0.00	0.00	0.00

3-1

Field name: 3-1

09/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 09/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	388.50 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.71	0.00	0.00	0.00	0.00

3-16

Field name: 3-16

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	63.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.50	0.00	0.00	0.00	0.00

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	180.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	0.00	0.00	0.00	0.00

3-17

Field name: 3-17

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	199.50 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.08	0.00	0.00	0.00	0.00

3-17

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	437.96 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.33	0.00	0.00	0.00	0.00

3-2

Field name: 3-2

09/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 09/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	419.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	16.00	132.80	0.00
Total actual harvest content	8.91	0.00	0.00	0.00	0.00

3-20

Field name: 3-20

3-20

09/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 09/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	378.72 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.53	0.00	0.00	0.00	0.00

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	706.72 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00
10/09/2023	158.67 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.91	0.00	0.00	0.00	0.00

3-21

Field name: 3-21

09/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 09/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	588.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.64	0.00	0.00	0.00	0.00

3-33

Field name: 3-33

09/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 09/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	336.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.76	0.00	0.00	0.00	0.00

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	422.07 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.83	0.00	0.00	0.00	0.00

3-8

Field name: 3-8

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	240.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	0.00	0.00	0.00	0.00

3-9

Field name: 3-9

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	506.08 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	38.93	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

1-1 - 09/18/2022: Oats, silage-soft dough

Field name: 1-1

Crop: Oats, silage-soft dough Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	61.31	25.16	159.81	2,618.68	30.00 *ton*
Application event totals		61.31	25.16	159.81	2,618.68	

1-1 - 05/23/2023: Corn, silage

Field name: 1-1

Crop: Corn, silage Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

1-1 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	583,924.99 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	875,887.48 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	583,924.99 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	875,887.48 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	583,924.99 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	875,887.48 *gal*
Application event totals		14.17	3.36	14.31	168.64	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

1-1 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,459,812.48 *gal*
Application event totals		0.52	0.00	0.00	37.42	

2-1 - 09/20/2022: Oats, silage-soft dough

Field name: 2-1

Crop: Oats, silage-soft dough Plant date: 09/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	83.45	34.24	217.52	3,564.32	35.00 *ton*
Application event totals		83.45	34.24	217.52	3,564.32	

2-13 - 09/17/2022: Oats, silage-soft dough

Field name: 2-13

Crop: Oats, silage-soft dough Plant date: 09/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2-13 - 09/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	160.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

2-31 - 09/16/2022: Oats, silage-soft dough

Field name: 2-31

Crop: Oats, silage-soft dough

Plant date: 09/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	30.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

2-9 - 09/20/2022: Oats, silage-soft dough

Field name: 2-9

Crop: Oats, silage-soft dough

Plant date: 09/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	36.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

2-9 - 05/22/2023: Corn, silage

Field name: 2-9

Crop: Corn, silage

Plant date: 05/22/2023

2-9 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	254.34	104.36	662.92	10,862.68	160.00 *ton*
Application event totals		254.34	104.36	662.92	10,862.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	750,760.70 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,126,141.06 *gal*
Application event totals		14.17	3.36	14.31	168.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	750,760.70 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,126,141.06 *gal*
Application event totals		14.17	3.36	14.31	168.64	

2-9 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	750,760.70 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,126,141.06 *gal*
Application event totals		14.17	3.36	14.31	168.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-1 - 09/15/2022: Oats, silage-soft dough

Field name: 3-1

Crop: Oats, silage-soft dough

Plant date: 09/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3-1 - 09/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	80.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

3-16 - 09/18/2022: Oats, silage-soft dough

Field name: 3-16

Crop: Oats, silage-soft dough Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	12.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

3-16 - 05/23/2023: Corn, silage

Field name: 3-16

Crop: Corn, silage Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	286.13	117.41	745.78	12,220.52	60.00 *ton*
Application event totals		286.13	117.41	745.78	12,220.52	

3-16 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	250,253.56 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	375,380.35 *gal*
Application event totals		14.17	3.36	14.31	168.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	250,253.56 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	375,380.35 *gal*
Application event totals		14.17	3.36	14.31	168.64	

3-16 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	250,253.56 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	375,380.35 *gal*
Application event totals		14.17	3.36	14.31	168.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	625,633.92 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-17 - 09/18/2022: Oats, silage-soft dough

Field name: 3-17

Crop: Oats, silage-soft dough

Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3-17 - 09/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	55.64	22.83	145.01	2,376.21	35.00 *ton*
Application event totals		55.64	22.83	145.01	2,376.21	

3-17 - 05/20/2023: Corn, silage

Field name: 3-17

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	317.92	130.45	828.65	13,578.36	200.00 *ton*
Application event totals		317.92	130.45	828.65	13,578.36	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-17 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	750,760.70 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,126,141.06 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	750,760.70 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,126,141.06 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	750,760.70 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,126,141.06 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-17 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,876,901.76 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-2 - 09/15/2022: Oats, silage-soft dough

Field name: 3-2

Crop: Oats, silage-soft dough Plant date: 09/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	94.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

3-20 - 09/16/2022: Oats, silage-soft dough

Field name: 3-20

Crop: Oats, silage-soft dough Plant date: 09/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3-20 - 09/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	56.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

3-20 - 05/20/2023: Corn, silage

Field name: 3-20

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	1,167,849.98 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,751,774.98 *gal*
Application event totals		14.17	3.36	14.31	168.64	

3-20 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	1,167,849.98 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,751,774.98 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	1,167,849.98 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,751,774.98 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	2,919,624.96 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-21 - 09/17/2022: Oats, silage-soft dough

Field name: 3-21

Crop: Oats, silage-soft dough

Plant date: 09/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	122.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

3-33 - 09/15/2022: Oats, silage-soft dough

Field name: 3-33

Crop: Oats, silage-soft dough

Plant date: 09/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	57.23	23.48	149.16	2,444.10	34.00 *ton*
Application event totals		57.23	23.48	149.16	2,444.10	

3-33 - 05/21/2023: Corn, silage

Field name: 3-33

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	269.30	110.50	701.91	11,501.67	160.00 *ton*
Application event totals		269.30	110.50	701.91	11,501.67	

3-33 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	709,051.77 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,063,577.66 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	709,051.77 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,063,577.66 *gal*
Application event totals		14.17	3.36	14.31	168.64	

3-33 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	709,051.77 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	1,063,577.66 *gal*
Application event totals		14.17	3.36	14.31	168.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,772,629.44 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-8 - 05/23/2023: Corn, silage

Field name: 3-8

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3-8 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	321.90	132.08	839.01	13,748.09	90.00 *ton*
Application event totals		321.90	132.08	839.01	13,748.09	

05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	333,671.42 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	500,507.13 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	333,671.42 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	500,507.13 *gal*
Application event totals		14.17	3.36	14.31	168.64	

3-8 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	333,671.42 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	500,507.13 *gal*
Application event totals		14.17	3.36	14.31	168.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	834,178.56 *gal*
Application event totals		0.52	0.00	0.00	37.42	

3-9 - 05/21/2023: Corn, silage

Field name: 3-9

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3-9 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	264.12	108.37	688.42	11,280.48	120.00 *ton*
Application event totals		264.12	108.37	688.42	11,280.48	

05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	542,216.06 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	813,324.09 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	542,216.06 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	813,324.09 *gal*
Application event totals		14.17	3.36	14.31	168.64	

3-9 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.85	3.36	14.31	146.19	542,216.06 *gal*
Canal Water	Surface water	0.31	0.00	0.00	22.45	813,324.09 *gal*
Application event totals		14.17	3.36	14.31	168.64	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

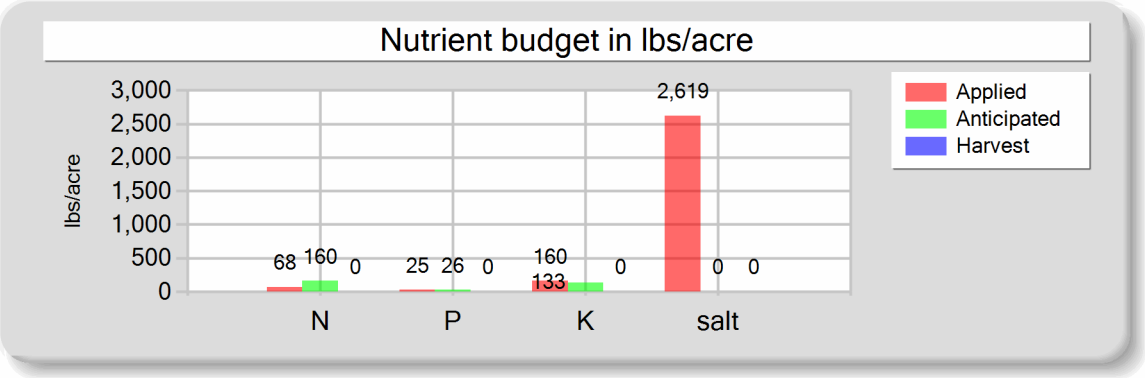
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.52	0.00	0.00	37.42	1,355,540.16 *gal*
Application event totals		0.52	0.00	0.00	37.42	

B. NUTRIENT BUDGET

1-1 - 09/18/2022: Oats, silage-soft dough

Field name: 1-1 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	61.31	25.16	159.81	2,618.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	68.31	25.16	159.81	2,618.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	68.31	25.16	159.81	2,618.68
Applied to removed ratio	0.00	0.00	0.00	0.00

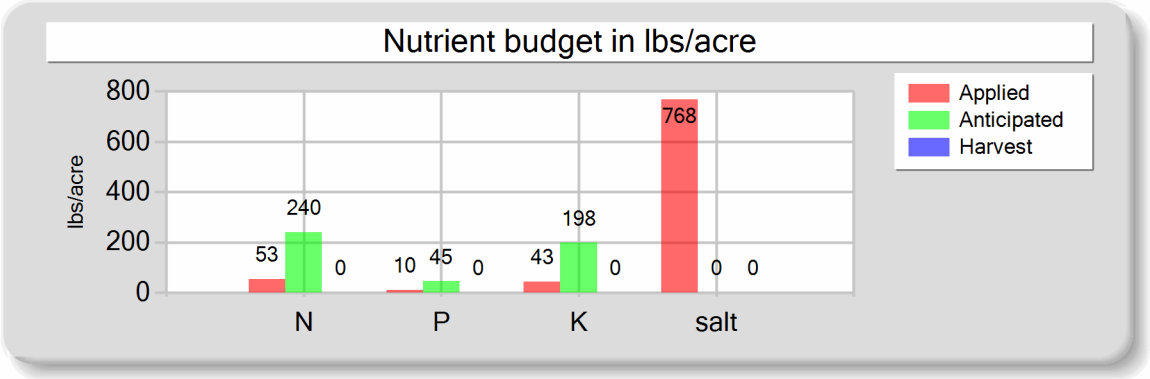
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

1-1 - 05/23/2023: Corn, silage

Field name: 1-1 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	53.15	10.09	42.92	767.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	53.15	10.09	42.92	767.82
Applied to removed ratio	0.00	0.00	0.00	0.00

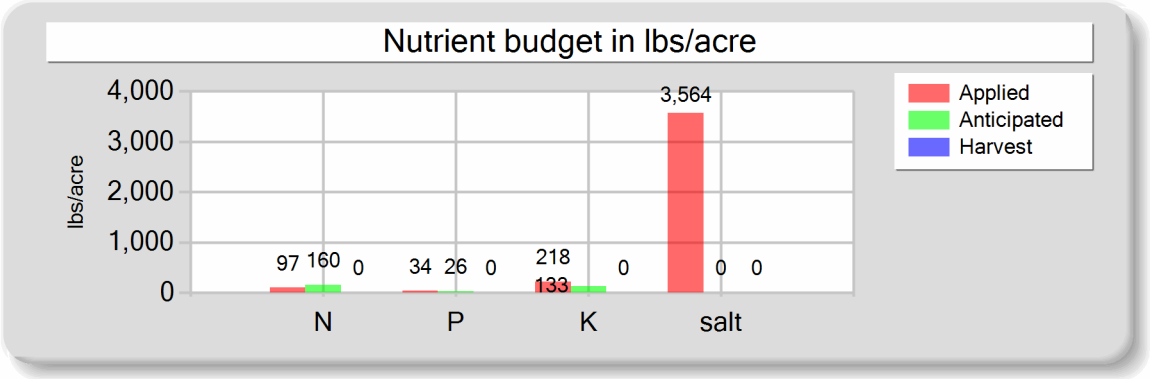
Fresh water applied	
12,846,349.80	*gallons*
473.09	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
1,751,774.97	*gallons*
64.51	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-1 - 09/20/2022: Oats, silage-soft dough

Field name: 2-1 Crop: Oats, silage-soft dough Plant date: 09/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	83.45	34.24	217.52	3,564.32
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	97.45	34.24	217.52	3,564.32
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	97.45	34.24	217.52	3,564.32
Applied to removed ratio	0.00	0.00	0.00	0.00

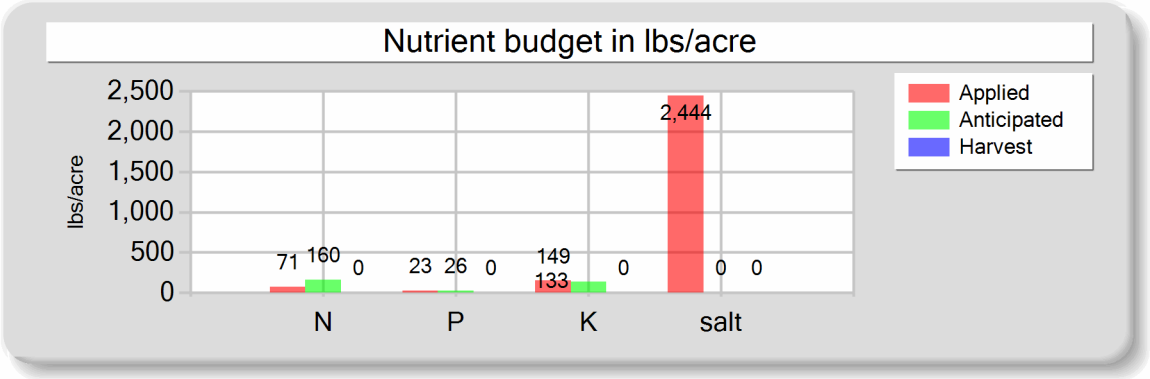
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-13 - 09/17/2022: Oats, silage-soft dough

Field name: 2-13 Crop: Oats, silage-soft dough Plant date: 09/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	71.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	71.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

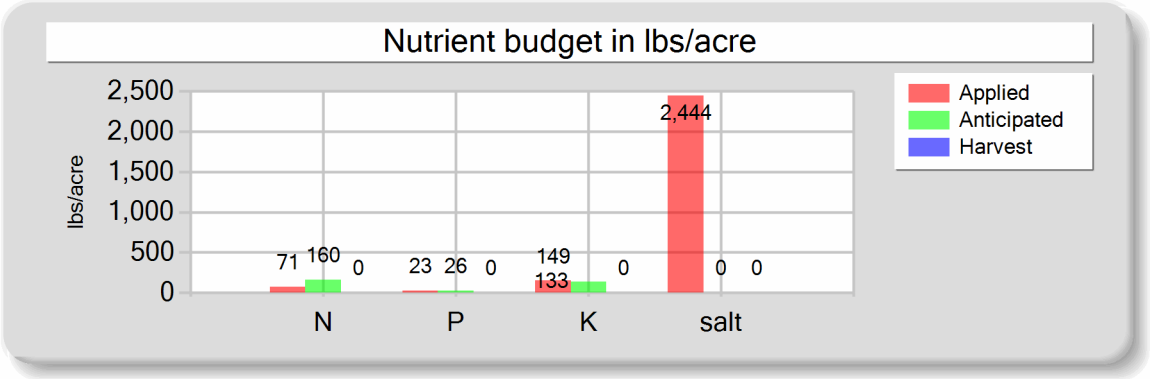
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-31 - 09/16/2022: Oats, silage-soft dough

Field name: 2-31 Crop: Oats, silage-soft dough Plant date: 09/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	71.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	71.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

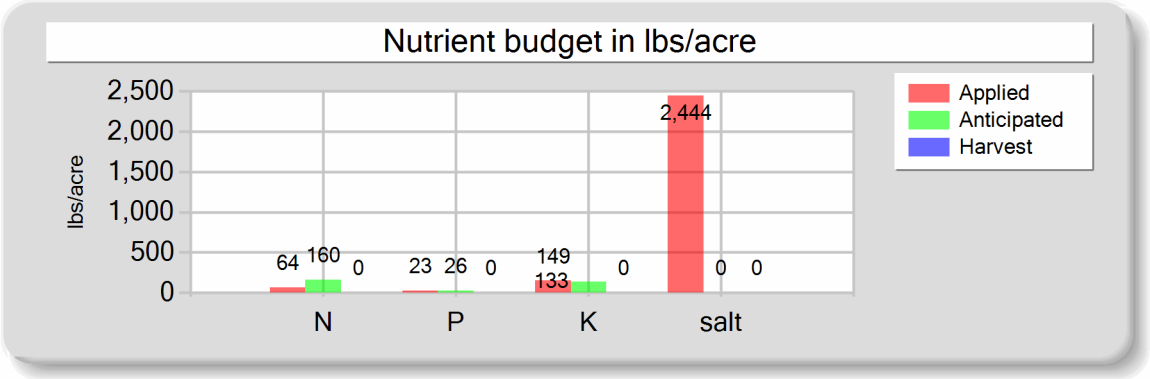
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-9 - 09/20/2022: Oats, silage-soft dough

Field name: 2-9 Crop: Oats, silage-soft dough Plant date: 09/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	64.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	64.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

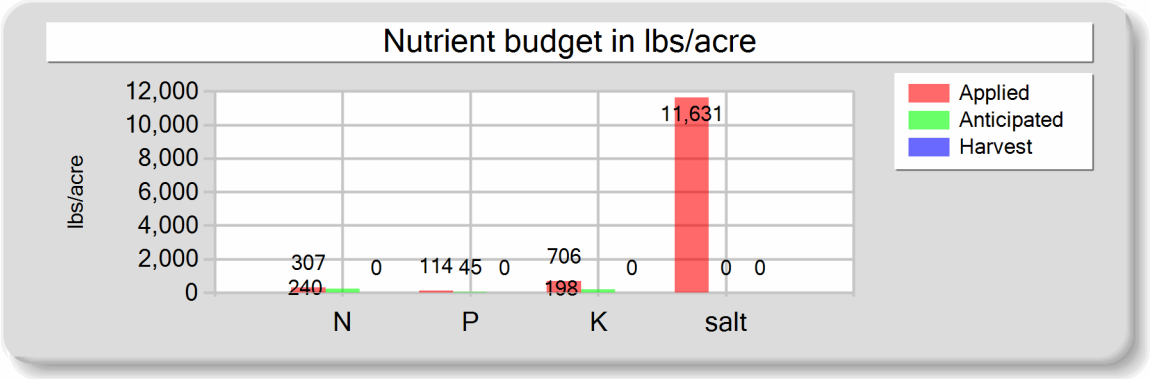
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-9 - 05/22/2023: Corn, silage

Field name: 2-9 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	254.34	104.36	662.92	10,862.68
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.49	114.45	705.84	11,630.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	307.49	114.45	705.84	11,630.51
Applied to removed ratio	0.00	0.00	0.00	0.00

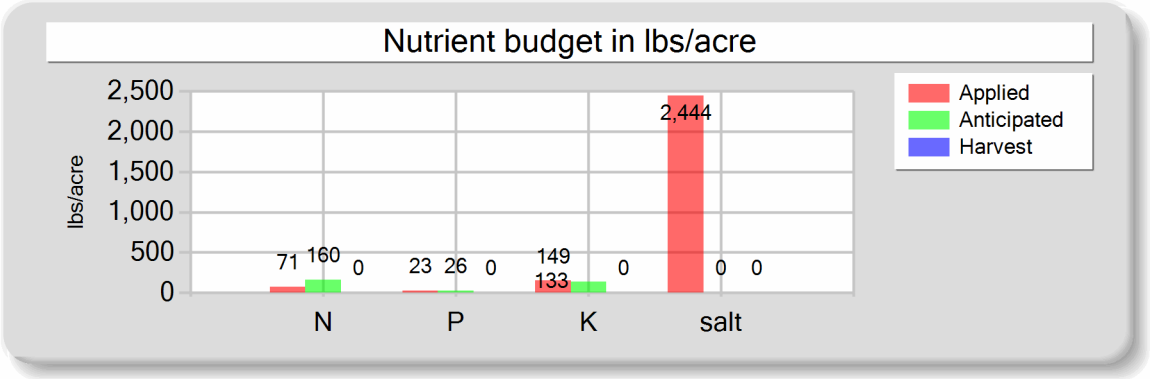
Fresh water applied	
16,516,735.50	*gallons*
608.26	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
2,252,282.10	*gallons*
82.94	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

3-1 - 09/15/2022: Oats, silage-soft dough

Field name: 3-1 Crop: Oats, silage-soft dough Plant date: 09/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	71.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	71.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

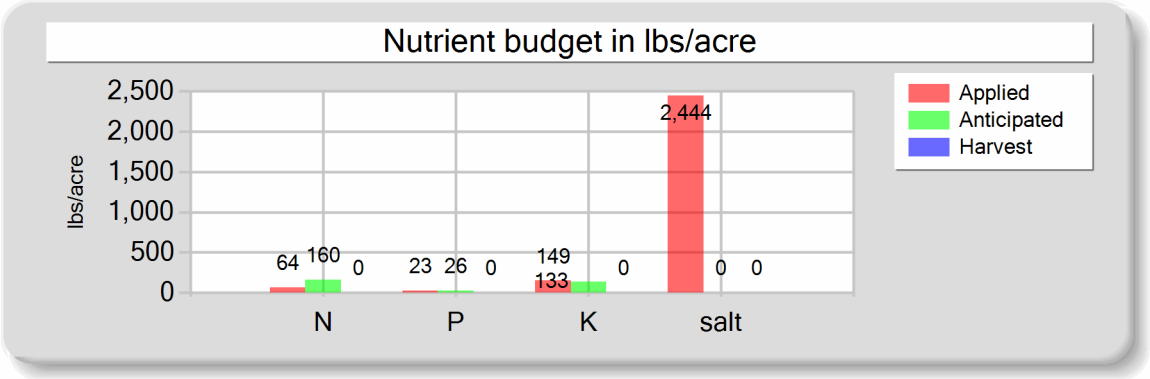
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-16 - 09/18/2022: Oats, silage-soft dough

Field name: 3-16 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	64.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	64.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

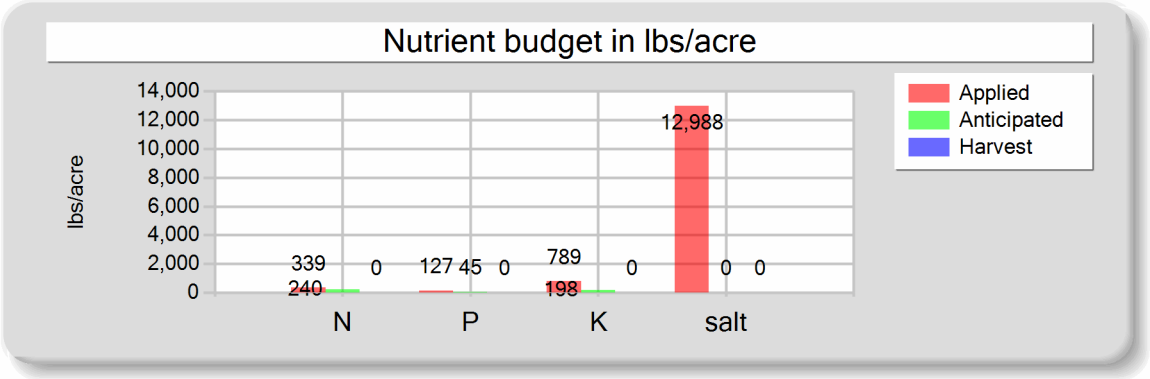
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-16 - 05/23/2023: Corn, silage

Field name: 3-16 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	286.13	117.41	745.78	12,220.52
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	339.28	127.49	788.70	12,988.34
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	339.28	127.49	788.70	12,988.34
Applied to removed ratio	0.00	0.00	0.00	0.00

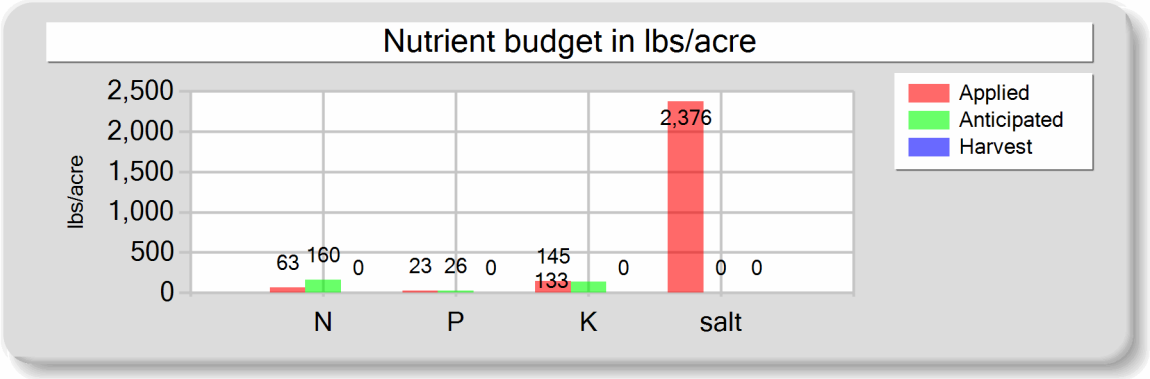
Fresh water applied	
5,505,578.49	*gallons*
202.75	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
750,760.68	*gallons*
27.65	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

3-17 - 09/18/2022: Oats, silage-soft dough

Field name: 3-17 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	55.64	22.83	145.01	2,376.21
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	62.64	22.83	145.01	2,376.21
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	62.64	22.83	145.01	2,376.21
Applied to removed ratio	0.00	0.00	0.00	0.00

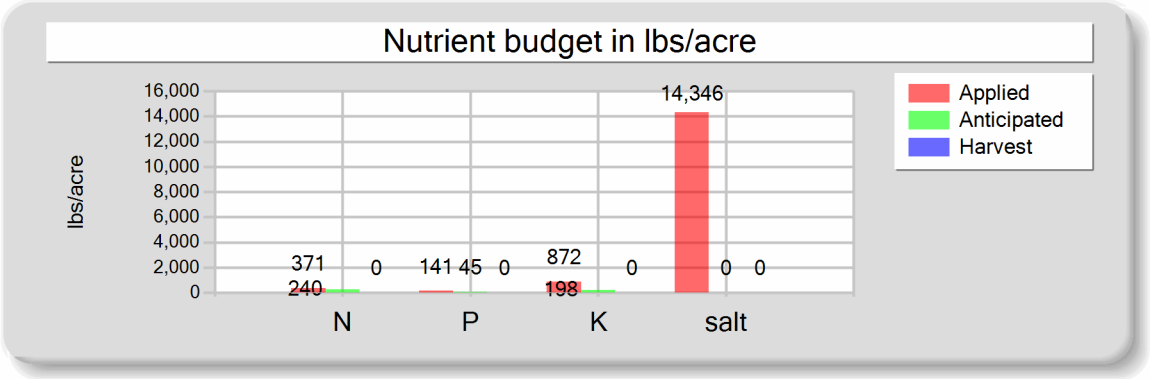
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-17 - 05/20/2023: Corn, silage

Field name: 3-17 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	317.92	130.45	828.65	13,578.36
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	371.08	140.54	871.56	14,346.18
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	371.08	140.54	871.56	14,346.18
Applied to removed ratio	0.00	0.00	0.00	0.00

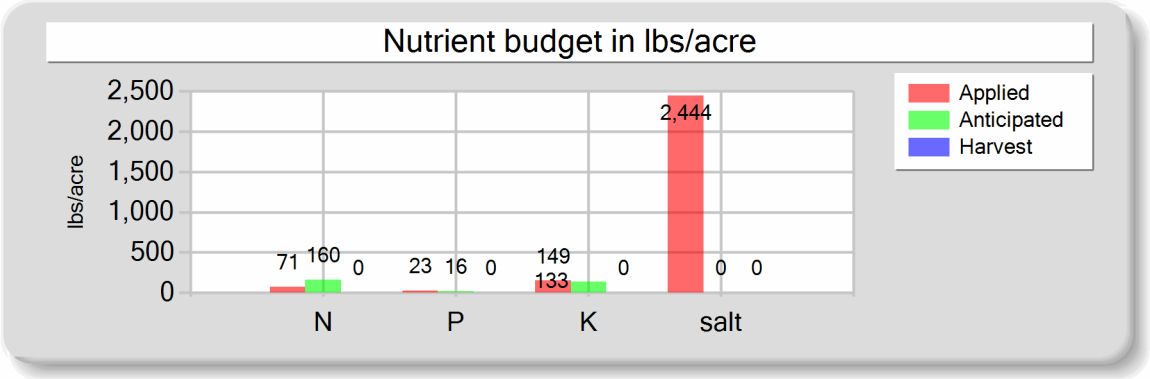
Fresh water applied	
16,516,735.50	*gallons*
608.26	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
2,252,282.10	*gallons*
82.94	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

3-2 - 09/15/2022: Oats, silage-soft dough

Field name: 3-2 Crop: Oats, silage-soft dough Plant date: 09/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	71.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	16.00	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	71.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

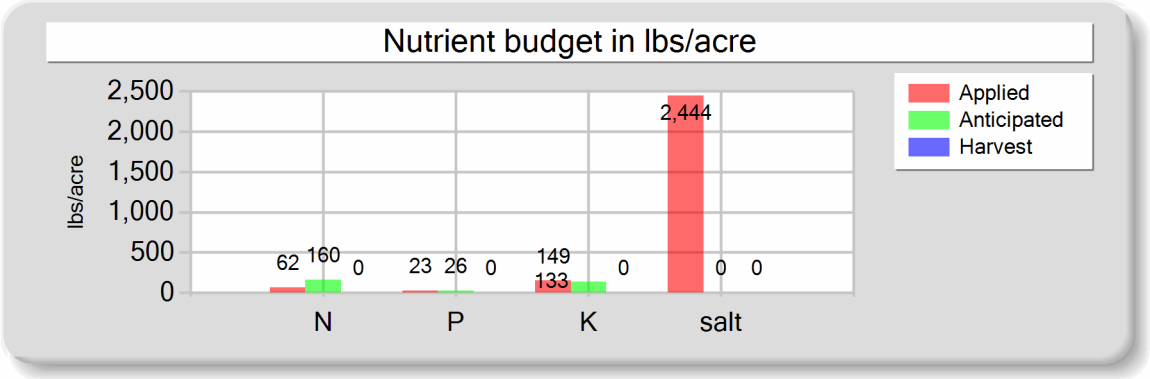
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-20 - 09/16/2022: Oats, silage-soft dough

Field name: 3-20 Crop: Oats, silage-soft dough Plant date: 09/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	61.89	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	61.89	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

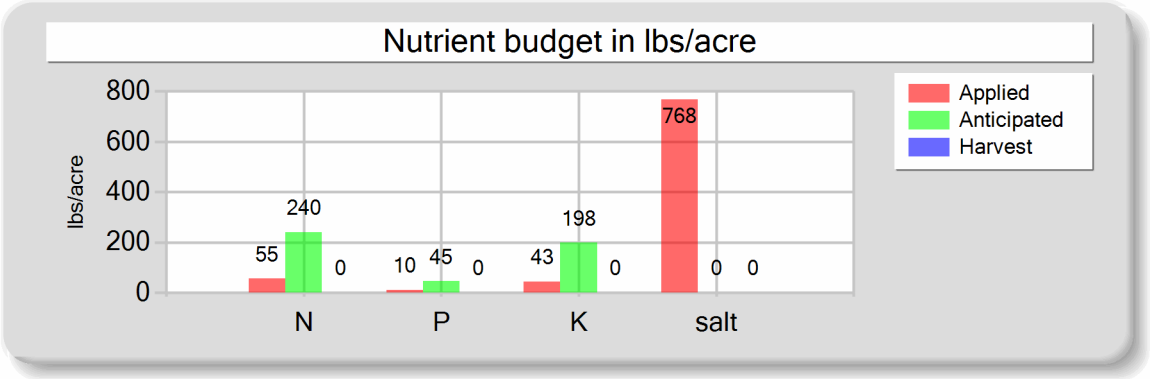
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-20 - 05/20/2023: Corn, silage

Field name: 3-20 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	55.49	10.09	42.92	767.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	55.49	10.09	42.92	767.82
Applied to removed ratio	0.00	0.00	0.00	0.00

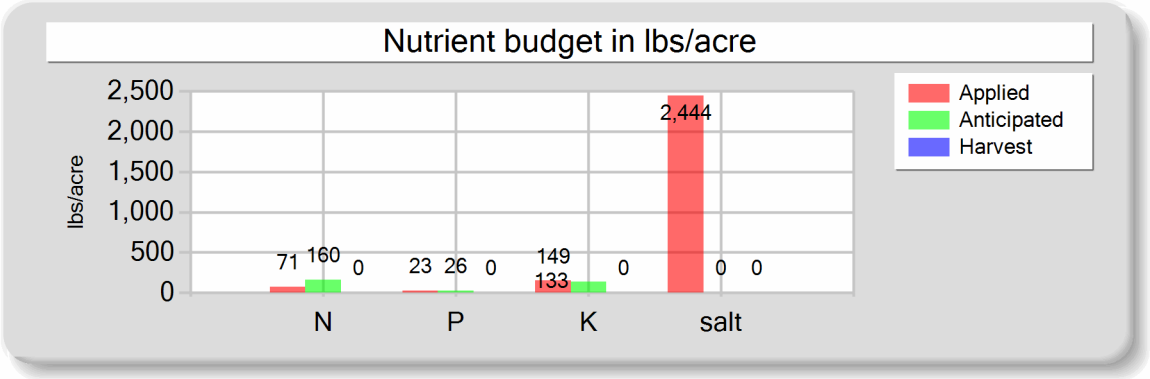
Fresh water applied	
25,692,699.66	*gallons*
946.17	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
3,503,549.94	*gallons*
129.02	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
2	*harvests*

3-21 - 09/17/2022: Oats, silage-soft dough

Field name: 3-21 Crop: Oats, silage-soft dough Plant date: 09/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	71.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	71.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

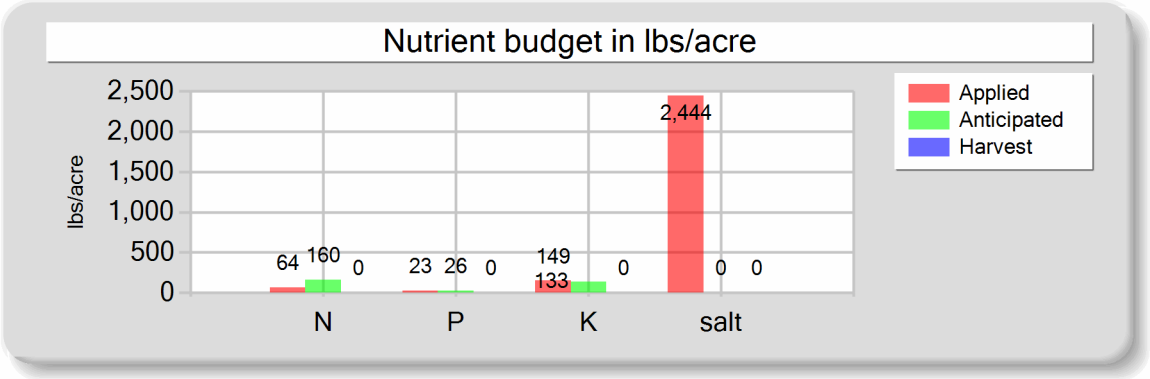
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-33 - 09/15/2022: Oats, silage-soft dough

Field name: 3-33 Crop: Oats, silage-soft dough Plant date: 09/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.23	23.48	149.16	2,444.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	64.23	23.48	149.16	2,444.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	64.23	23.48	149.16	2,444.10
Applied to removed ratio	0.00	0.00	0.00	0.00

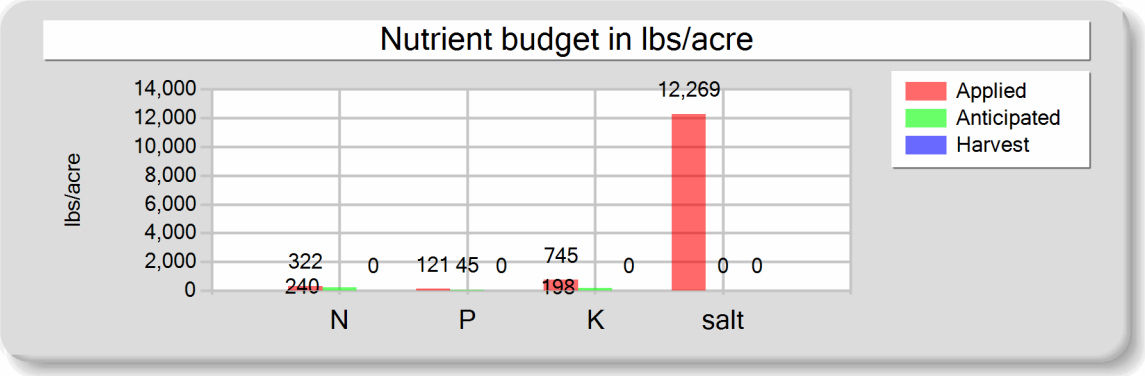
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3-33 - 05/21/2023: Corn, silage

Field name: 3-33 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	269.30	110.50	701.91	11,501.67
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.45	120.59	744.83	12,269.49
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	322.45	120.59	744.83	12,269.49
Applied to removed ratio	0.00	0.00	0.00	0.00

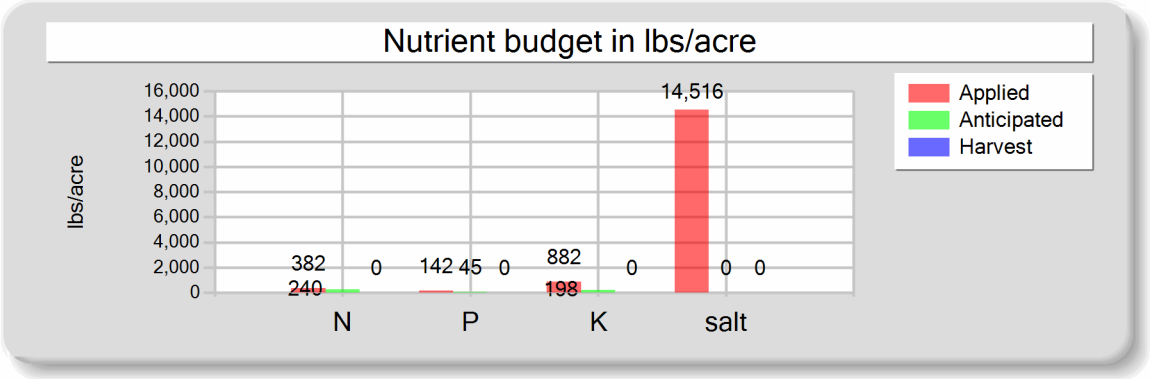
Fresh water applied	
15,599,139.06	*gallons*
574.46	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
2,127,155.31	*gallons*
78.34	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

3-8 - 05/23/2023: Corn, silage

Field name: 3-8 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	321.90	132.08	839.01	13,748.09
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	382.05	142.17	881.92	14,515.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	382.05	142.17	881.92	14,515.91
Applied to removed ratio	0.00	0.00	0.00	0.00

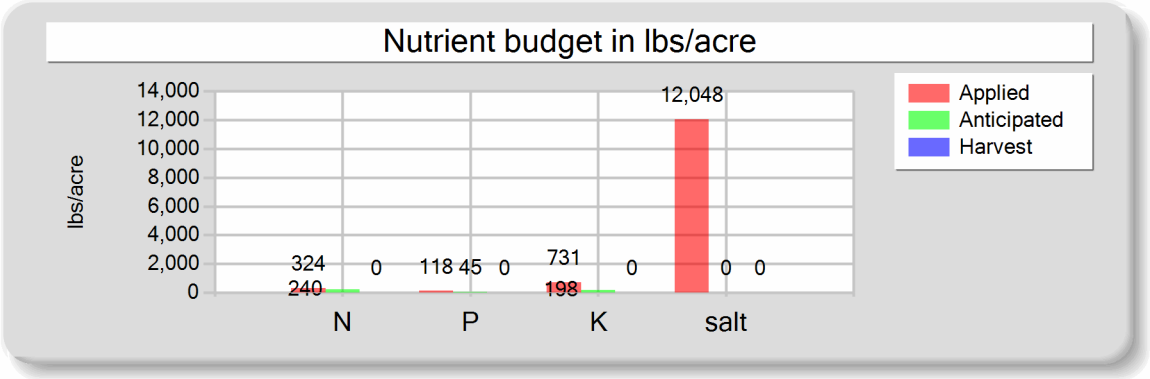
Fresh water applied	
7,340,771.31	*gallons*
270.34	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
1,001,014.26	*gallons*
36.86	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

3-9 - 05/21/2023: Corn, silage

Field name: 3-9 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	264.12	108.37	688.42	11,280.48
Process wastewater	41.56	10.09	42.92	438.56
Fresh water	4.59	0.00	0.00	329.27
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	324.27	118.46	731.33	12,048.30
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	324.27	118.46	731.33	12,048.30
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
11,928,753.39	*gallons*
439.30	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
1,626,648.18	*gallons*
59.90	*acre-inches*
4.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2091506-01

Sample and source description: Manure 2091506-01

Sample date: 09/15/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	6,360.00	40,400.00	13,000.00	6,000.00	8,000.00	2,890.00	5,490.00		66.20
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302050-01

Sample and source description: Lagoon Water 2302050-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	248.00	248.00			65.60	587.00								3,170.00	3,000
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303401-01

Sample and source description: Lagoon Water 2303401-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	305.00	305.00			56.70	473.00								3,740.00	2,300
DL	0.60	5.00			0.10	10.00								1.00	1

Lagoon Water 2307239-01

Sample and source description: Lagoon Water 2307239-01

Sample date: 07/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	39.80	39.80			9.66	41.10								956.00	420
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	103.00	97.80			16.80	42.50								1,430.00	500
DL	0.60	0.60			0.10	1.00								1.00	1

C. FRESH WATER ANALYSES

Canal Water

2310023-01

Sample description: 2310023-01

Sample date: 09/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.60										25.70	43
DL	0.40										1.00	1

Domestic Well/ #1 5106 N Quincy

Domestic Well/ #1 5106 N Quincy

2312292-01

Sample description: 2312292-01

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			7.72								410.00	
DL			0.40								1.00	

Domestic Well/ #11 Sperry's

2312292-04

Sample description: 2312292-04

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											346.00	
DL											1.00	

Domestic Well/ #2 Paul 5200 N Quincy

2312292-02

Sample description: 2312292-02

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			6.54								376.00	
DL			0.40								1.00	

Domestic Well/ #27 Hyline

Domestic Well/ #27 Hyline

2312292-06

Sample description: 2312292-06

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			3.85								283.00	
DL			0.40								1.00	

Domestic Well/ #5 Nutchers Dairy

2312292-03

Sample description: 2312292-03

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			10.30								471.00	
DL			0.40								1.00	

Domestic Well/ 16 Home Dairy

2312292-05

Sample description: 2312292-05

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			6.31								395.00	
DL			0.40								1.00	

Domestic Well/ 2519 Barn Hart Rd

Domestic Well/ 2519 Barn Hart Rd

2312292-08

Sample description: 2312292-08

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			11.00								472.00	
DL			0.40								1.00	

Domestic Well/ 5818 S Gratton Rd

2312292-07

Sample description: 2312292-07

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			4.53								421.00	
DL			0.40								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

No plant tissue analyses entered.

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

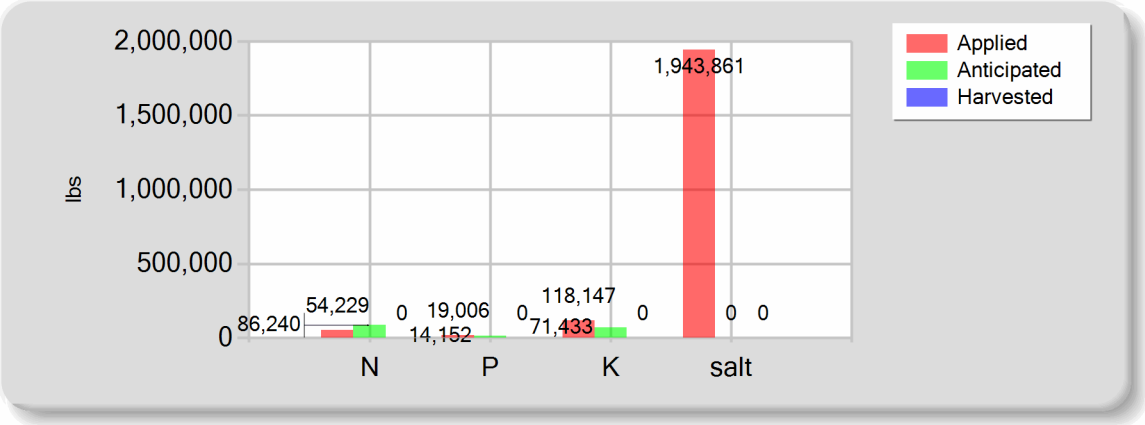
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

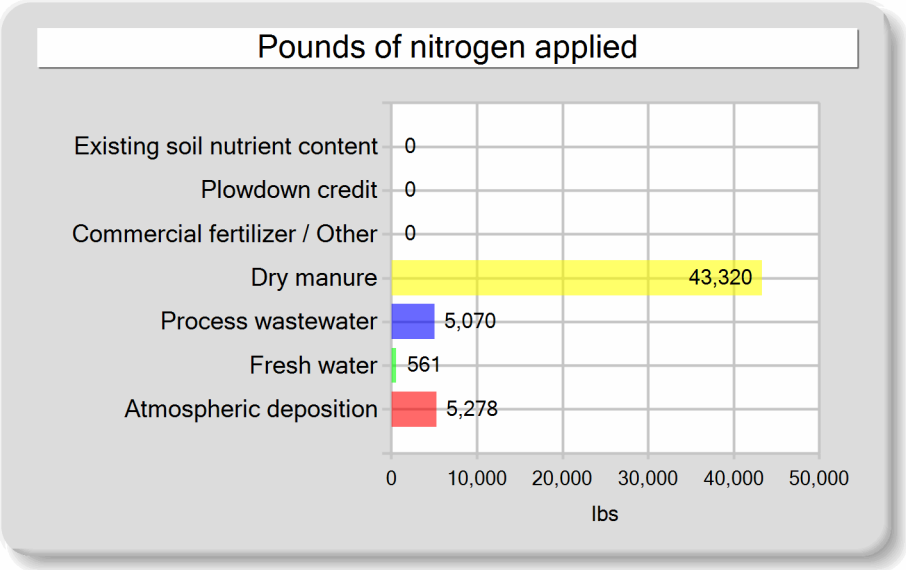
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

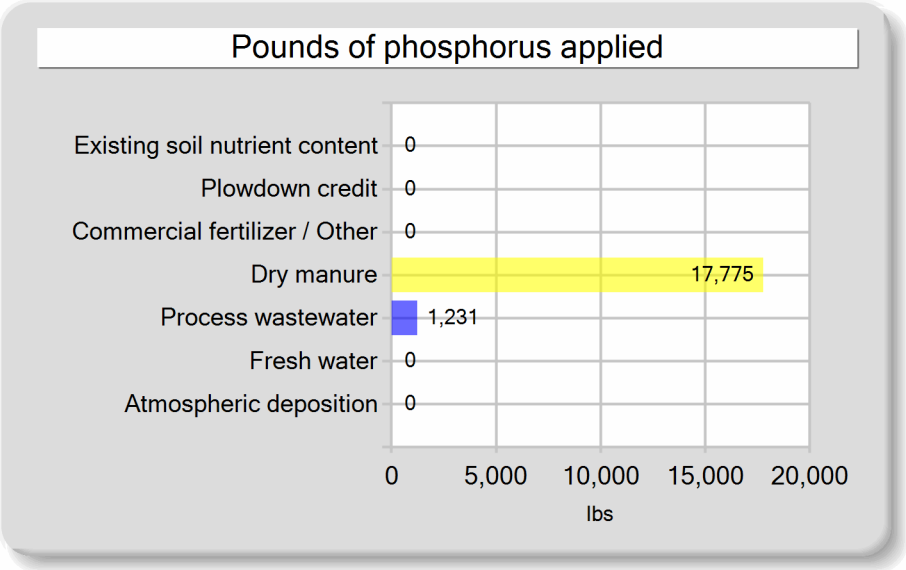
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	43,320.08	17,775.21	112,911.70	1,850,186.73
Process wastewater	5,070.13	1,230.59	5,235.74	53,503.94
Fresh water	560.52	0.00	0.00	40,170.42
Atmospheric deposition	5,278.00	0.00	0.00	0.00
Total nutrients applied	54,228.73	19,005.80	118,147.44	1,943,861.08
Anticipated crop nutrient removal	86,240.00	14,152.40	71,432.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	54,228.73	19,005.80	118,147.44	1,943,861.08
Applied to removed ratio	0.00	0.00	0.00	0.00

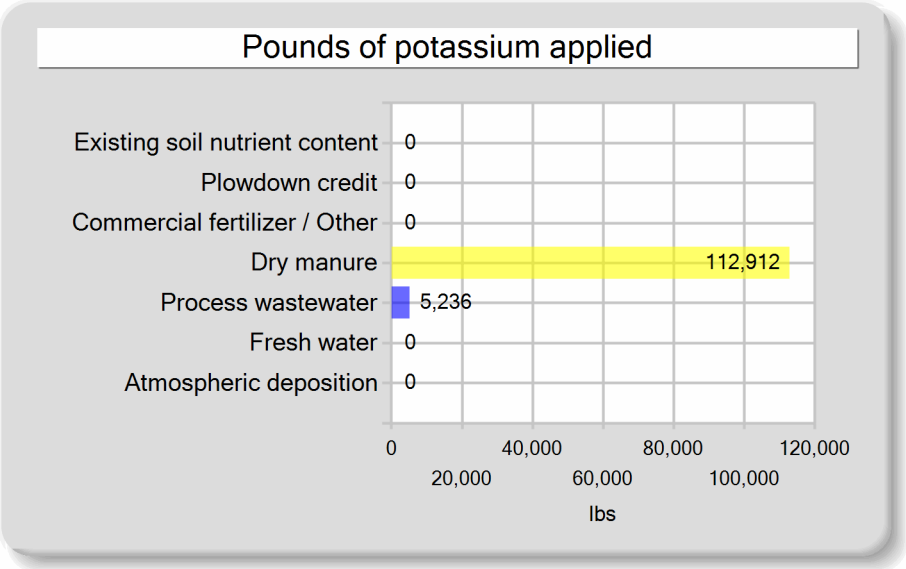
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

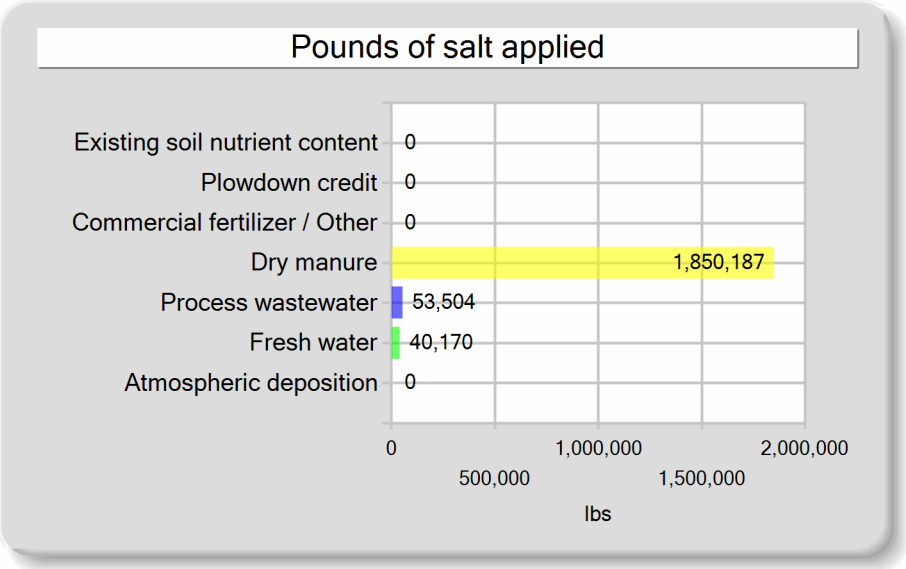


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*The Owner/Operator sold cows in March 2023, they will continue to comply with the Dairy General Order requirements.

*Winter Forage & Corn Silage were not sample in 2023, we will be sure to sample in 2024.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Edwin H. Smith and Sons	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Rick Smith 7-11-24	7-11-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	